Football Match Prediction: Techniques, Data, and Research Gaps

Abstract

Football match outcome prediction has become a prominent application of machine learning because the sport combines rich historical data with high outcome uncertainty. This literature review synthesizes the evidence from the provided papers on machine learning methods for football and soccer match prediction, focusing on algorithms, datasets, features, evaluation metrics, limitations, and future research directions. The reviewed literature shows that no single algorithm is consistently best across all settings; instead, performance depends strongly on the choice of features, the type of dataset, the league or competition studied, and the evaluation metric used. Tree-based ensembles, logistic regression, support vector machines, and neural models all appear in the literature, with gradient-boosted trees and Random Forest often reported as strong performers in soccer-specific tasks. Common predictors include team strength measures, home advantage, betting odds, goal-based statistics, FIFA or Elo-derived ratings, and player- or team-level variables. The literature also repeatedly highlights the need for benchmark datasets, better interpretability, richer event and tracking data, and more consistent evaluation protocols.pubmed.ncbi.nlm.nih+2

Keywords: football prediction, soccer analytics, machine learning, random forest, XGBoost, logistic regression, neural networks, systematic review

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1. INTRODUCTION

Predicting football match outcomes is challenging because results depend on many interacting factors, including team strength, tactical context, player availability, home advantage, and stochastic match events. The supplied literature shows that machine learning has become the main analytical approach for this task, but studies differ widely in their datasets, feature engineering choices, and evaluation practices. As a result, claims that one algorithm is universally superior are not supported by the evidence.mendel-journal+2

This review synthesizes the literature thematically rather than paper by paper. It asks which algorithms are most common, which features are most used, which datasets dominate the field, which metrics are preferred, where studies agree or disagree, and what gaps remain. The goal is to produce a publication-ready synthesis that can support future research in football match outcome prediction.arxiv+1

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2. RESEARCH QUESTIONS

This review is organized around the following research questions:

- **RQ1:** Which machine learning algorithms are most commonly used in football match outcome prediction?
- **RQ2:** Which features are most frequently used, and which appear most useful?
- **RQ3:** Which datasets, leagues, and competitions dominate the literature?
- **RQ4:** Which evaluation metrics are preferred?
- **RQ5:** What limitations, disagreements, and research gaps recur across studies?

These questions reflect the thematic focus of the supplied review papers, which emphasize algorithms, data, feature engineering, evaluation, and future research directions.pubmed.ncbi.nlm.nih+1

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3. REVIEW METHODOLOGY

3.1 REVIEW DESIGN

This manuscript is a literature review based exclusively on the papers provided by the user. No outside claims are introduced beyond the content of those papers. The structure follows the same logic used in the soccer systematic review and the broader team-sport review, both of which emphasize transparent study selection, thematic grouping, and comparison across methods and datasets.arxiv+1

3.2 SCOPE

The review focuses on machine learning techniques for football or soccer match outcome prediction. The evidence base includes soccer-focused review papers and an English Premier League prediction study, along with a broader team-sport review that includes soccer as well as other sports. The manuscript does not claim to be a full database-backed systematic review unless the exact search process and counts are independently documented.mendel-journal+2

3.3 STUDY SELECTION LOGIC

The supplied soccer systematic review reports PRISMA-based screening and indicates that studies were identified, screened, assessed for full-text eligibility, and then included for analysis. That paper also states that 145 studies were initially identified and 32 were fully reviewed, with outcome measures extracted and analyzed. This manuscript uses the same logic of transparent selection, but because it is built from the user-provided corpus, it reports only what can be verified from those papers.pubmed.ncbi.nlm.nih

3.4 DATA EXTRACTION FRAMEWORK

For thematic synthesis, the following variables are extracted from the provided papers:

Authors	Year	League / Competition	Dataset	Features	Algorithm	Metric	Best Result	Limitations

Bunker, Yeung, Fujii	2024	Soccer	Multiple datasets across the literature	Goals, ratings, player/team info, event/tracking data	CatBoost, Random Forest, deep learning	Accuracy, ranked probability score	CatBoost strong on goal-only datasets	Need broader model comparison and interpretability arxiv
Choi, Foo, Chua	2023	English Premier League	EPL 2022–2023 season	Balanced sampling features	Random Forest, Logistic Regression, Linear SVC, XGBoost	Accuracy, betting profit	Random Forest best accuracy; Logistic Regression best betting profit	Single-league scope; metric dependence mendel-journal

Rico-González et al.	2022	Soccer	145 initially identified; 32 fully reviewed	Diverse ML applications in soccer	Various	Outcome measures extracted	Broad thematic synthesis	Comparability limited; benchmark scarcity pubmed.ncbi.nlm.nih

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4. RESULTS

4.1 ALGORITHMS STUDIED

The literature covers a broad set of machine learning algorithms, including logistic regression, Random Forest, support vector machines, XGBoost, CatBoost, artificial neural networks, and deep learning approaches. The broader team-sport review notes that artificial neural networks were common in early studies, but later work increasingly compared multiple model families instead of relying on a single method. The soccer chapter similarly highlights gradient-boosted tree models as strong performers in some datasets, especially when used with soccer-specific rating features.mendel-journal+2

The EPL study compared Random Forest, Logistic Regression, Linear Support Vector Classifier, and Extreme Gradient Boosting for binary and multiclass classification. It found that Random Forest achieved the highest accuracy for binary classification, whereas Logistic Regression produced the highest simulated betting profit. This distinction is important because it shows that the best model depends on whether the goal is pure classification or practical decision-making.mendel-journal

4.2 DATASETS AND COMPETITIONS

Soccer is the dominant sport in the supplied literature. The English Premier League appears prominently, and the literature also refers to broader soccer datasets, benchmark competition data, and multi-season league data. The systematic review confirms that soccer has a more developed ML evidence base than other team sports, although benchmark datasets remain limited overall.arxiv+2

The reviewed sources also indicate that different dataset types affect model performance. The soccer chapter notes that gradient-boosted trees such as CatBoost are currently reported as best-performing on datasets containing only goals as match features, but that deeper comparisons across richer datasets are still needed. The EPL study used data from the 2022–2023 season and tested the models under balanced and imbalanced sampling strategies.arxiv+1

4.3 FEATURES USED FOR PREDICTION

Feature engineering is one of the central themes in the literature. Common feature groups include team strength indicators, historical form, home advantage, betting odds, goals and shots, player statistics, FIFA-based ratings, and Elo-based variables. The soccer chapter specifically argues that new rating systems using both player- and team-level information deserve further investigation, and it also highlights the potential of spatiotemporal tracking and event data.mendel-journal+1

The literature suggests that betting odds are useful but not universally dominant. Some studies find that odds contribute meaningfully, while others show only marginal or statistically insignificant benefits when odds are combined with public match data. The Kelly Index study expanded the feature space using Elo-based variables and reported competitive performance, especially when matches were decomposed by predictive difficulty. Overall, the evidence suggests that feature quality and domain relevance are often more important than the raw number of features.pubmed.ncbi.nlm.nih+1

4.4 EVALUATION METRICS

Accuracy is the most frequently reported metric in the supplied literature, especially in classification-based comparisons. However, the studies also show that accuracy alone can be misleading. The EPL study demonstrates this clearly: Random Forest had the highest accuracy, but Logistic Regression produced the highest betting profit. This means that the metric chosen can change the interpretation of model quality.pubmed.ncbi.nlm.nih+1

The soccer chapter also discusses ranked probability score as a useful alternative for probabilistic forecasting. This is important because football outcome prediction often involves three classes or probability distributions rather than simple binary labels. The literature therefore supports the use of multiple evaluation metrics, especially when the practical objective is not just classification accuracy but calibrated forecasting or downstream utility.arxiv+1

4.5 AGREEMENTS ACROSS STUDIES

The literature agrees on several broad points. First, machine learning is viable for football outcome prediction, but performance varies widely by dataset and feature design. Second, no single algorithm is universally dominant. Third, richer and more context-aware features tend to improve predictive performance more reliably than raw algorithmic complexity.pubmed.ncbi.nlm.nih+2

There is also agreement that soccer is the most mature subdomain in terms of data availability and benchmarking, though even here the field lacks fully standardized datasets. Most studies also recognize that interpretable models and practical utility matter, not just raw accuracy.mendel-journal+2

4.6 DISAGREEMENTS ACROSS STUDIES

The most obvious disagreement concerns which model family performs best. The soccer chapter emphasizes gradient-boosted trees such as CatBoost on goal-only datasets. The EPL study, by contrast, finds that Random Forest is best for accuracy while Logistic Regression is best for betting profit. The broader review suggests that artificial neural networks were historically common, but that multiple model comparisons are preferable.arxiv+2

These disagreements are not truly contradictory because they arise from different datasets, feature sets, task formulations, and metrics. A model that performs best under binary classification may not be best under multiclass forecasting or profit-based evaluation. The literature therefore supports a conditional conclusion rather than a universal ranking.mendel-journal

5. THEMATIC SYNTHESIS

5.1 ALGORITHM SELECTION DEPENDS ON THE DATA

The strongest cross-study lesson is that algorithm choice cannot be separated from dataset design. The review literature repeatedly states that feature engineering and data representation are crucial. In soccer, models using ratings and goals-based features can perform very well, while in another setting odds or match statistics may be more influential.pubmed.ncbi.nlm.nih+2

5.2 FEATURE ENGINEERING IS CENTRAL

The supplied papers consistently show that carefully engineered inputs outperform naive feature sets. Team strength, recent form, home advantage, odds, goals, shots, Elo, FIFA ratings, and player-level variables all appear in the literature. The direction of the evidence suggests that better features often matter more than more complex algorithms.arxiv+2

5.3 BETTING UTILITY IS NOT THE SAME AS ACCURACY

A key insight from the EPL study is that the most accurate model is not necessarily the most useful for betting. This is a strong reminder that evaluation should match the intended application. For pure forecasting, accuracy or probabilistic scores may be appropriate; for betting or decision support, simulated profit or calibration may matter more.mendel-journal+1

5.4 BENCHMARKING IS STILL LIMITED

The literature repeatedly notes the lack of benchmark datasets, especially outside soccer. Even within soccer, dataset differences make comparisons difficult. This is one of the main reasons the literature does not support a definitive “best model” conclusion.pubmed.ncbi.nlm.nih+1

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6. TABLES FOR THE PAPER

TABLE 1. ALGORITHM COMPARISON ACROSS THE SUPPLIED STUDIES

Algorithm family	Studies using it	Reported strengths	Reported weaknesses
Logistic Regression	EPL study; broader football prediction studies mendel-journal	Interpretable; best betting profit in the EPL study mendel-journal	Not always top accuracy
Random Forest	EPL study; broader soccer review arxiv+1	High accuracy in binary EPL classification mendel-journal	Profit and accuracy may diverge
XGBoost / Gradient Boosting	EPL study; soccer review arxiv+1	Strong ensemble performance arxiv	Depends on features and dataset
CatBoost	Soccer chapter arxiv	Strong on goal-only datasets arxiv	Needs broader comparative testing
SVM / Linear SVC	EPL study; broader literature mendel-journal	Competitive baseline	Not consistently best

ANN / Deep Learning	Broader review and soccer chapter pubmed.ncbi.nlm.nih+1	Historically common; flexible nonlinear modeling	Interpretability concerns; mixed performance

TABLE 2. FEATURE TYPES AND THEIR ROLE

Feature group	Examples in the literature	Typical role
Team strength ratings	Elo, pi-ratings, FIFA ratings arxiv	Capture latent team quality
Match context	Home advantage, recent form, recovery time mendel-journal	Reflect situational effects
Betting market data	Betting odds and odds movement arxiv	Strong benchmark signal, sometimes difficult to beat
Match statistics	Goals, shots, possession, cards arxiv+1	Basic performance indicators
Player/team hybrids	Player-level and team-level information arxiv	Promising but underexplored
Event/tracking data	Spatiotemporal tracking, event data arxiv	Future research direction

TABLE 3. METRICS AND INTERPRETATION

Metric	Why it is used	Limitation
Accuracy	Simple and common mendel-journal	Can hide imbalance and calibration issues
F1-score	Helpful for class balance	Not always reported consistently
ROC-AUC	Useful for ranking ability	Less common in the supplied papers
Ranked probability score	Suitable for probabilistic outcomes arxiv	Less intuitive than accuracy
Betting profit	Measures practical utility mendel-journal	Depends on betting assumptions

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7. FIGURE PLACEHOLDERS

FIGURE 1. PRISMA FLOW DIAGRAM

A PRISMA-style diagram should show:

- Records identified.
- Duplicates removed.
- Records screened.
- Full texts assessed.
- Studies included.

The soccer systematic review demonstrates this reporting style and shows that 145 records were initially identified and 32 studies were fully reviewed.pubmed.ncbi.nlm.nih

FIGURE 2. ALGORITHM PERFORMANCE MAP

A bar chart or heatmap can compare:

- Random Forest.
- Logistic Regression.
- XGBoost.
- CatBoost.
- SVM.
- Neural Networks.

This would visually reinforce the paper’s main conclusion that performance is context dependent.arxiv+1

FIGURE 3. FEATURE IMPORTANCE CATEGORIES

A bar chart could show the relative prominence of:

- Ratings.
- Odds.
- Match statistics.

- Player information.
- Event/tracking data.

This would illustrate the central role of feature engineering in the literature.arxiv+1

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8. DISCUSSION

The reviewed literature makes one point especially clear: football match outcome prediction is not mainly a contest between competing algorithms, but a problem shaped by data quality, feature design, and evaluation choice. The strongest models in one study may not be strongest in another because the target, dataset, and objective differ. For example, Random Forest was most accurate in the EPL study, but Logistic Regression generated the best betting profit, showing that predictive accuracy and practical usefulness are not equivalent.mendel-journal+1

The literature also shows that soccer is the most developed subdomain, but even there the field lacks standardized benchmark datasets and consistent evaluation practices. This makes cross-study comparison difficult and limits the strength of broad claims. The evidence therefore favors a cautious interpretation: ensemble methods often perform well, but no model can yet be declared universally superior.pubmed.ncbi.nlm.nih+1

Another important implication is that future progress will likely come from richer data rather than only newer algorithms. The soccer chapter specifically recommends better rating systems that combine player- and team-level information, plus event and tracking data, which could capture tactical structure more effectively than traditional box-score variables alone. That recommendation is strongly aligned with the broader review's conclusion that feature engineering is often more important than increasing instance count or model complexity.arxiv+1

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9. LIMITATIONS OF THE REVIEW

This manuscript is limited by the fact that it uses only the papers provided by the user. It therefore cannot claim to be a full database-driven systematic review unless the search process is separately conducted and documented. It also cannot provide exact corpus-wide counts such as “X of Y studies used Random Forest” unless those numbers are extracted from a complete dataset of included papers. The literature supports such quantitative synthesis, but only when the study set is explicitly defined and counted.pubmed.ncbi.nlm.nih+1

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10. FUTURE RESEARCH DIRECTIONS

The literature supports several future directions:

- Standardized benchmark datasets.
- Cross-league evaluation.
- Women’s football prediction.
- Explainable AI methods.
- Player tracking and event data.
- Graph neural networks.
- Transformer-based models.
- Hybrid rating systems.
- Calibration of probabilistic predictions.

These directions are consistent with the soccer chapter’s emphasis on richer player/team representations, spatiotemporal data, and improved interpretability. They also align with the broader review’s call for better benchmark datasets and more comprehensive model comparison.arxiv+1

11. CONCLUSION

The supplied literature shows that machine learning has become a central tool for football match outcome prediction, but it does not support a single universally best algorithm. Instead, performance depends on the interaction between algorithm choice, feature engineering, dataset structure, and evaluation metric. Tree-based ensembles, logistic regression, support vector machines, and neural methods all have roles in the literature, but their relative effectiveness changes across leagues, competitions, and outcome formulations. The most important next steps are richer data, standardized benchmark datasets, stronger evaluation protocols, and more interpretable models rather than incremental algorithmic variation alone.mendel-journal+2

REFERENCES

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